

Ellipsoid Method and the Amazing Oracles

Separation oracles, parallel cuts, and the practice of convex optimization

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Section 1

Agenda



Agenda



Part 1 — Foundations why the ellipsoid method? · separation oracles · cut types · termination



Part 2 — Amazing Oracles robust optimization · parametric networks · matrix inequalities



Part 3 — Ellipsoid Revisited search space · deep & central cuts · parallel cuts · stable factorization



Part 4 — Applications & Practice FIR design · discrete optimization · implementation · benchmarks

Section 2

Foundations



Why the Ellipsoid Method?

- The interior-point method evaluates **all** constraints; the ellipsoid method needs only a **separation oracle** 🔍
- Naturally suited to a **moderate number of variables** with a **huge or infinite** number of constraints ∞
- It cannot exploit **sparsity** — but the oracle can exploit **structure** 🧩
- One algorithm serves feasibility, optimization, discrete design, and bisection 🎯
The oracle, not the ellipsoid, does the intellectual work. 🧠

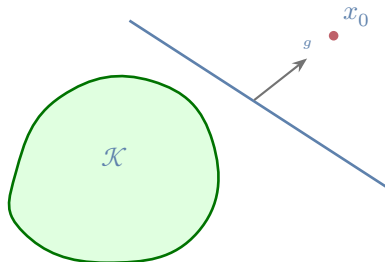


Separation Oracle: the Real Star

When queried at $x_0 \in \mathbb{R}^n$, the oracle Ω answers one of two ways:

- ① “ $x_0 \in \mathcal{K}$ ” ☒ — stop, the point is feasible, or
- ② a hyperplane separating x_0 from $\mathcal{K}$:

$$g^\top(x - x_0) + \beta \leq 0, \quad \beta \geq 0, \quad g \neq 0, \quad \forall x \in \mathcal{K}.$$





Three Flavours of Cutting Planes

cut		condition	geometry
central	○	$\beta = 0$	the plane passes through x_0
deep	▼	$\beta > 0$	x_0 is strictly cut off
shadow	◐	$\beta < 0$	x_0 already lies inside

- A convex f gives a cut for free: $(g, \beta) = (\partial f(x_0), f(x_0))$ ✨
- Differentiable $f \rightarrow$ the subgradient is the gradient.



From Feasibility to Optimization

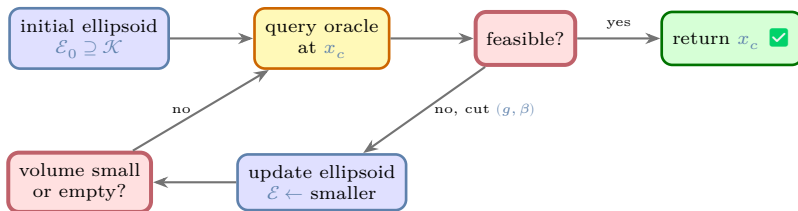
- Add a level constraint $f_0(x) \leq \gamma$ and shrink the **sublevel set** as soon as a better point is found:

$$\begin{array}{ll} \text{minimize} & \gamma, \\ \text{subject to} & \Phi(x, \gamma) \leq 0, \ x \in \mathcal{K}. \end{array}$$

- A **binary search** on γ is just a nested feasibility problem 
- no solution $\Rightarrow$ infeasible; no effect $\Rightarrow$ the cut was too shallow.



The Cutting-plane Loop





Termination and the Stall Guard

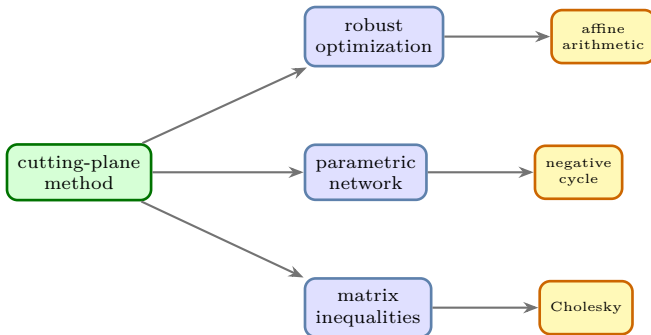
- Comparing a **scale-dependent width** against an **absolute** tolerance can never succeed ●
 - Once the bracket is at machine resolution, the midpoint is the only representable point
 - The counter pins and inner subproblems are solved for nothing ⌚
 - **Remedy:** terminate when the midpoint is no longer strictly inside the bracket — a *stall guard*, not a tighter tolerance 🛡️
- An absolute stopping threshold should always be paired with a stall test.* 📌

Section 3

Amazing Oracles



Three Oracles, One Method





Robust Convex Optimization

- The worst case over an uncertainty set $\mathcal{Q}$ — a **robust counterpart** stays convex, but its constraints become infinite ∞

$$\begin{array}{ll} \text{minimize} & \sup_{q \in \mathcal{Q}} f_0(x, q), \\ \text{subject to} & f_j(x, q) \leq 0, \quad \forall q \in \mathcal{Q}, \quad j = 1, \dots, m. \end{array}$$

- Oracle: maximize over q and return the cut at the maximizer
- **Affine arithmetic** propagates uncertainty symbolically, avoiding the 2^K vertices of a box
- Certified feasible iff the affine upper bound $\bar{f}_j(x) \leq 0$ for every j .



Parametric Network Problems

- A directed graph $G = (V, E)$ with $u_i - u_j \leq h_{ij}(x, \gamma)$
- Feasible **iff** G has **no negative cycle** — the oracle is negative-cycle detection
- Equivalently: $W_k(x, \gamma) = \sum_{(i,j) \in C_k} h_{ij}(x, \gamma) \geq 0$ for every cycle C_k
- Fast detectors: Bellman–Ford, Tarjan (1976), Howard (policy iteration) ⚡
- Application: optimal **matrix scaling** under the min–max–ratio criterion ⚖️



Matrix Inequalities and SDP

- Seek x with $F(x) \succeq 0$, where $F(x) = F_0 + x_1 F_1 + \cdots + x_n F_n$
- $A \succeq 0 \iff v^\top A v \geq 0$ for all v
- **Cholesky / $LDL^\top$ witness:** if the factorization fails at row p , then

$$v = R_{:,p}^{-1} e_p, \quad v^\top F_{:,p}(x_0) v < 0,$$

$$\text{cut} = (-v^\top \partial F_{:,p} v, -v^\top F_{:,p} v).$$

- Examples: **minimum eigenvalue** (EVP) and matrix-norm minimization 



Estimating a Correlation Function

- The Gaussian criterion combines a **convex** trace term with a **concave** log-determinant — a difference of convex functions
- It is convex exactly on the trust region $0 \prec \Omega \preceq 2Y$
- Beyond it, apply the **convex–concave procedure (CCP)**: majorize the concave part by its tangent, minimize the convex surrogate, repeat
- Do **not** impose $0 \preceq \Omega \preceq 2Y$ as a hard constraint — it biases the estimate

Section 4

Ellipsoid Revisited

● The Ellipsoid as a Search Space

- The search space must be cheap to store, cheap to update, and guaranteed to contain $\mathcal{K}$ — the ellipsoid is all three ●

$$\mathcal{E} = \{ x \mid (x - x_c)^\top Q^{-1} (x - x_c) \leq \kappa \}$$

- $O(n^2)$ parameters · closed-form update · containment by induction 📺
- Eigenvectors give the axes, eigenvalues the lengths; the volume is

$$\text{vol}(\mathcal{E}) = \kappa^{n/2} \sqrt{\det Q} \text{vol}(\mathcal{B}^n).$$



Updating the Ellipsoid (Deep Cut)

- Let $\tilde{g} = Qg$, $\omega = g^T \tilde{g}$, and $\tau = \sqrt{\kappa \omega}$ 🔧

$$x_c^+ = x_c - \frac{\rho}{\omega} \tilde{g}, \quad Q^+ = Q - \frac{\sigma}{\omega} \tilde{g} \tilde{g}^T, \quad \kappa^+ = \delta \kappa.$$

- Deep cut** ($\beta > 0$): $\rho = \frac{\tau + n\beta}{n+1}$, $\sigma = \frac{2\rho}{\tau + \beta}$, $\delta = \frac{n^2}{n^2 - 1} \cdot \frac{\tau^2 - \beta^2}{\tau^2}$
- Empty if $\beta > \tau$; too shallow if $n\beta < -\tau$ ⚠

○ Central Cut

- The special case $\beta = 0$ is much simpler — worth its own implementation ○

$$\rho = \frac{\tau}{n+1}, \quad \sigma = \frac{2}{n+1}, \quad \delta = \frac{n^2}{n^2-1}.$$

- Same update map, cheaper coefficients 🎯



Parallel Cuts

- The oracle returns a **pair** of cuts sharing a normal g :

$$g^\top(x - x_c) + \beta_1 \leq 0, \quad g^\top(x - x_c) + \beta_2 \geq 0, \quad \forall x \in \mathcal{K}.$$

- Produced by any two-sided linear constraint $l \leq a^\top x + b \leq u$ 
- The update removes a **slab**, not a half-space $\rightarrow$ faster convergence 

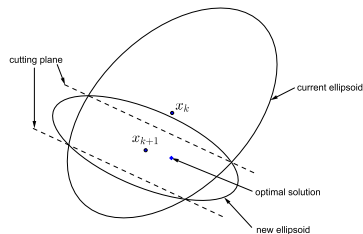


Figure 1: Parallel cuts



Parallel-cut Parameters

With $\tilde{g} = Qg$, $\tau^2 = \kappa\omega$, and the auxiliary quantities

$$\zeta_0 = \tau^2 - \beta_0^2, \quad \zeta_1 = \tau^2 - \beta_1^2, \quad \xi = \sqrt{\zeta_0 \zeta_1 + \left(\frac{n}{2}(\beta_1^2 - \beta_0^2)\right)^2},$$

the scalar triple is

$$\sigma = \frac{2\eta}{\tau^2 + \beta_0\beta_1 + \frac{n}{2}(\beta_0 + \beta_1)^2 + \xi}, \quad \rho = \sigma \cdot \frac{\beta_0 + \beta_1}{2},$$

$$\delta = \frac{n^2}{(n^2 - 1)\tau^2} \left(\frac{\zeta_0 + \zeta_1}{2} + \frac{\xi}{n} \right), \quad \eta = \tau^2 + n\beta_0\beta_1.$$

This form stays finite when $\beta_0 + \beta_1 = 0$ — the symmetric case a mean-offset formula cannot handle. ❧



How Fast Does the Volume Shrink?

- Apply the matrix-determinant lemma to the rank-one downdate:

$$\det Q^+ = (1 - \sigma) \det Q \quad \implies \quad \frac{\text{vol}(\mathcal{E}^+)}{\text{vol}(\mathcal{E})} \leq e^{-1/(2n)}.$$

- After k iterations the volume is at most $e^{-k/(2n)}$ of the initial volume 
- To reach a fraction ϵ of it: $k \approx 2n \ln(1/\epsilon)$ iterations 



Numerical Drift and the Stable LDL^T Variant

- Repeated rank-one downdates can lose positive definiteness — **silently** 🤖
- Keep a factored shape instead:

$$Q = \kappa L D L^T, \quad w = L^{-1}g, \quad z = D^{-1}w, \quad \omega = w^T z, \quad q = L^{-T}z.$$

- Three triangular sweeps + a factor update; positive definiteness **by construction** 🛡️
- Compiled cost is comparable to the direct form — near-free insurance ✅

Section 5

Applications



FIR Filter Design

- Magnitude constraints $L(\omega) \leq |H(\omega)| \leq U(\omega)$ are **not** convex in $\mathbf{h}$ ❌
- **Spectral factorization** rewrites them convexly in the autocorrelation $\mathbf{r}$ ✨

$$L^2(\omega) \leq R(\omega) \leq U^2(\omega), \quad R(\omega) = |H(\omega)|^2.$$

- Two-sided bounds $\Rightarrow$ **parallel cuts** $\Rightarrow$ markedly fewer iterations 🚀

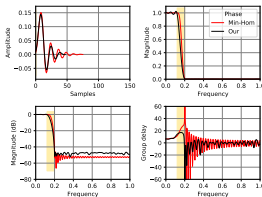


Figure 2: Lowpass result



The Multiplierless FIR Pipeline



- A CSD coefficient is a signed sum of powers of two $\rightarrow$ shifts and adds, **no multipliers**

12
34

- The non-convex quantization constraint is folded **into the oracle** 🎯

 Spectral Factorization: FFT vs Roots

	FFT (Kolmogorov)	root-finding (Aberth–Ehrlich)
transform needed	FFT library	none
memory	$O(\text{oversample} \cdot N)$	$O(N)$
tuning	none	tolerance
round-trip error	$\sim 10^{-5}$	$\sim 10^{-3}$
use for	production, high order	exploration, moderate order

- Both return a valid minimum-phase factor — they differ in the numerical profile, not the mathematics 
- The optimizer may run fewer iterations with the better-conditioned root-based factor



Discrete Optimization

- Some variables live in a discrete set $\mathbb{D}$ — mixed-integer convex programming 🎲
- Relaxation + branch-and-bound **ignores convexity** ✖
- The oracle looks for a nearby discrete point x_d and cuts there:

$$g^\top(x - x_d) + \beta \leq 0, \quad \beta \geq 0, \quad g \neq 0.$$

- The cut can be **shallow**; on “no effect”, perturb and **retry** 🔁

	quantize-after	discrete in oracle
constraint	after optimization	inside the oracle
domain	$\mathbb{R}^N$	CSD-realizable
guarantee	none	at sampled frequencies
cost	single pass	multiple passes with retries



Maximum Likelihood and CCP

- The criterion $\log \det V + \text{Tr}(V^{-1}Y)$ is convex on $0 \preceq V \preceq 2Y$ ✓
- Beyond the region, **CCP** turns one non-convex problem into a sequence of convex ones

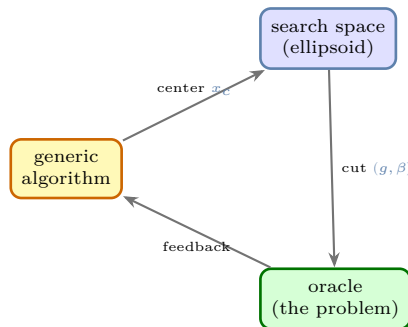
- A triangular factor must be inverted by back-substitution — a symmetric routine silently corrupts the gradient ⚠

Section 6

Implementation & Practice



Architecture: Algorithm · SearchSpace · Oracle



- One problem-agnostic loop serves **feasibility**, **optimization**, **quantized**, and **bisection** oracles 🧩
- Swap the search space (dense · factored · interval) without touching the algorithm 🛠️



Performance Lessons

- **Profile first:** in an interpreted oracle, the *number of calls* dominated, not the arithmetic 🔍
- Replace per-constraint callbacks with **one matrix-vector product + masks**, preserving the round-robin cursor 🎯
- **Per-iteration allocation is the tax** — pre-allocate and reuse scratch buffers 🗑️
- Early exit can beat vectorization in compiled code — **port the measurement, not the conclusion** 📌
- An unchanged iteration count on fixed input is strong evidence that a rewrite preserved behaviour ✅



Ellipsoid vs Interior-Point

problem	interior-point	ellipsoid (interp.)	ellipsoid (compiled)
Chebyshev center, $n = 10$	1.0×	2.9×	0.03×
min-eigenvalue LMI, $m = 8$	1.0×	14.6×	0.08×
FIR lowpass, $n = 48$	1.0×	7.7×	0.26×

- Runtimes relative to the interior-point solver; **below 1.0 is faster**
- The per-iteration constant (~language overhead) can flip the ranking
- Use the ellipsoid method when constraints are reachable **only through an oracle**

Section 7

Closing



Key Takeaways

The idea 💡

- the **oracle** is the intellectual centre — the ellipsoid is bookkeeping 🧠
- **parallel cuts** pay off whenever bounds are two-sided 📊
- discrete and robust problems fit the same loop 🎯

The practice 🛠️

- factor the ellipsoid (LDL^T) for **free insurance** 🛡️
- profile, pre-allocate, and re-measure 📈
- pick the method by **constraint access**, not by reputation 🌟



References

- Boyd & Vandenberghe, *Convex Optimization* — interior-point methods 
- Bland, Goldfarb & Todd (1981) — the ellipsoid method: a survey 
- Wu et al. (1999) — FIR design via spectral decomposition 
- Goodman (1997) — spectral factorization for FIR design 
- Liu et al. (2007) — affine arithmetic for robust geometric programming 
*Full citations and the complete argument live in the paper “**Ellipsoid Method and the Amazing Oracles.**”* 



Q&A

Ellipsoid Method and the Amazing Oracles

Questions? Discussion? 



Thank You

The oracle does the thinking; the ellipsoid just shrinks. 🟠 ▼

Slides built with Beamer · TikZ 🟢 · LuaLaTeX 📐 · Nord 🌙